

PRAKTIKUM SISTEM OPERASI

MODUL 13

PERINTAH DASAR LINUX



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

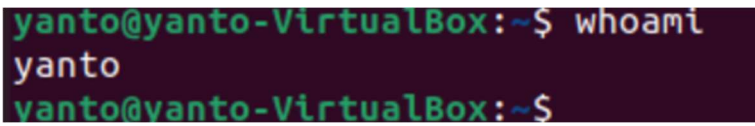
UNIVERSITAS TELKOM

2026

JURNAL/TP

1. Soal 1

a. Screenshot terminal



```
yanto@yanto-VirtualBox:~$ whoami
yanto
yanto@yanto-VirtualBox:~$
```

b. Jelaskan arti dari command prompt milik Anda!

Command prompt:

yanto@yanto-VirtualBox:~\$

memiliki arti:

- **yanto** = username yang sedang digunakan.
- **yanto-VirtualBox** = hostname atau nama komputer.
- **~** = direktori home user, yaitu /home/yanto.
- **\$** = menunjukkan bahwa user yang sedang aktif adalah user biasa (non-root).

Jika menggunakan user root, simbol \$ akan berubah menjadi #.

Perintah

whoami

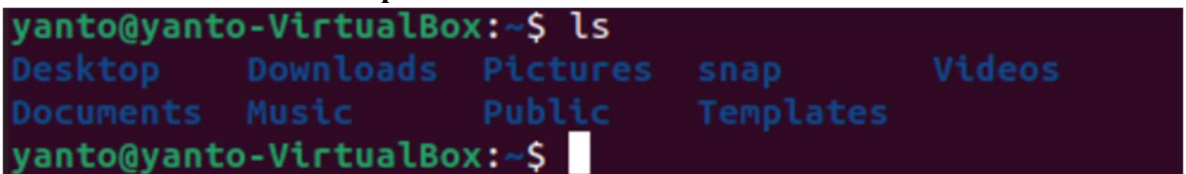
menghasilkan:

yanto

yang menunjukkan bahwa user yang sedang login adalah **yanto**.

2. Soal 2

a. Jalankan dan screenshot perintah berikut ini: ls



```
yanto@yanto-VirtualBox:~$ ls
Desktop    Downloads  Pictures   snap       Videos
Documents  Music      Public     Templates
yanto@yanto-VirtualBox:~$
```

b. Apakah option dan parameter dari perintah di atas?

Nama perintah = ls

Option = tidak ada

Parameter = tidak ada

c. Apa fungsi dari perintah tersebut?

Menampilkan isi direktori aktif.

d. Jalankan perintah berikut ini: `ls -al /`

```
yanto@yanto-VirtualBox:~$ ls -al
total 80
drwxr-x--- 16 yanto yanto 4096 Jun  7 17:37 .
drwxr-xr-x  3 root  root  4096 Mei 19 12:22 ..
-rw-----  1 yanto yanto   25 Jun  7 17:37 .bash_history
-rw-r--r--  1 yanto yanto  220 Mei 19 12:22 .bash_logout
-rw-r--r--  1 yanto yanto 3771 Mei 19 12:22 .bashrc
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .cache
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .config
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Desktop
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Documents
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Downloads
drwx-----  2 yanto yanto 4096 Jun 18 06:13 .gnupg
drwx-----  3 yanto yanto 4096 Mei 19 12:51 .local
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Music
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Pictures
-rw-r--r--  1 yanto yanto  807 Mei 19 12:22 .profile
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Public
drwx-----  4 yanto yanto 4096 Mei 19 12:51 snap
drwx-----  2 yanto yanto 4096 Jun  7 17:01 .ssh
-rw-r--r--  1 yanto yanto    0 Jun  2 11:08 .sudo_as_admin_successful
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Templates
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Videos
```

e. Apakah option dan parameter dari perintah di atas?

Nama perintah = `ls`

Option = `-al`

Parameter = `/`

f. Apa fungsi dari perintah tersebut?

Menampilkan seluruh isi direktori root secara detail termasuk hidden file.

g. Jelaskan mengapa perintah pada a dan e mempunyai hasil yang berbeda!

Karena:

- `ls` hanya menampilkan isi direktori saat ini.
- `ls -al /` menampilkan isi direktori root (`/`) secara detail dan juga file tersembunyi.

3. Pohon File

```
yanto@yanto-VirtualBox:~$ pwd
/home/yanto
yanto@yanto-VirtualBox:~$
```

b. Option dan parameter

Tidak ada option maupun parameter.

c. Fungsi

Menampilkan direktori kerja saat ini (working directory).

4. Perpindahan

```
yanto@yanto-VirtualBox:~$ ls
Desktop    Downloads  Pictures   snap       Videos
Documents  Music      Public     Templates
yanto@yanto-VirtualBox:~$ cd Public
yanto@yanto-VirtualBox:~/Public$
```

b. Option dan parameter

- Option = tidak ada
- Parameter = /

c. Fungsi

- Berpindah ke direktori root.

5. Direktori khusus

a. cd / cd ~

```
yanto@yanto-VirtualBox:~$ cd Public
yanto@yanto-VirtualBox:~/Public$ pwd
/home/yanto/Public
yanto@yanto-VirtualBox:~/Public$ cd ~
yanto@yanto-VirtualBox:~$ pwd
/home/yanto
yanto@yanto-VirtualBox:~$
```

Penjelasan

- cd / berpindah ke direktori root.
- cd ~ berpindah ke home directory user.

b. Cd ..

```
yanto@yanto-VirtualBox:~$ cd Templates
yanto@yanto-VirtualBox:~/Templates$ pwd
/home/yanto/Templates
yanto@yanto-VirtualBox:~/Templates$ cd Templates
bash: cd: Templates: No such file or directory
yanto@yanto-VirtualBox:~/Templates$ cd ..
yanto@yanto-VirtualBox:~$ pwd
/home/yanto
yanto@yanto-VirtualBox:~$
```

6. CRUD File

a. Copy file /proc/cpuinfo ke folder home

```
yanto@yanto-VirtualBox:~/Documents$ cp /proc/cpuinfo /home/yanto/
yanto@yanto-VirtualBox:~/Documents$ ls
File_Test.txt
yanto@yanto-VirtualBox:~/Documents$
```

- b. Membuktikan file telah tercopy

```
yanto@yanto-VirtualBox:~/Documents$ ls /home/yanto
cpuinfo  Documents  File_Test.txt  Pictures  snap        Videos
Desktop  Downloads  Music          Public    Templates
yanto@yanto-VirtualBox:~/Documents$
```

- c. Copy file /proc/uptime ke folder home

```
yanto@yanto-VirtualBox:~/Documents$ cp /proc/uptime /home/yanto
yanto@yanto-VirtualBox:~/Documents$ ls
File_Test.txt
yanto@yanto-VirtualBox:~/Documents$ ls -al /home/yanto
total 92
drwxr-x--- 16 yanto yanto 4096 Jun 18 07:00 .
drwxr-xr-x  3 root  root  4096 Mei 19 12:22 ..
-rw-----  1 yanto yanto   25 Jun  7 17:37 .bash_history
-rw-r--r--  1 yanto yanto  220 Mei 19 12:22 .bash_logout
-rw-r--r--  1 yanto yanto 3771 Mei 19 12:22 .bashrc
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .cache
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .config
-r--r--r--  1 yanto yanto 3500 Jun 18 06:58 cpuinfo
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Desktop
drwxr-xr-x  2 yanto yanto 4096 Jun 18 06:50 Documents
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Downloads
-rw-rw-r--  1 yanto yanto   32 Jun 18 06:53 File_Test.txt
drwx-----  2 yanto yanto 4096 Jun 18 06:13 .gnupg
drwx-----  3 yanto yanto 4096 Mei 19 12:51 .local
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Music
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Pictures
-rw-r--r--  1 yanto yanto  807 Mei 19 12:22 .profile
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Public
drwx-----  4 yanto yanto 4096 Mei 19 12:51 snap
drwx-----  2 yanto yanto 4096 Jun  7 17:01 .ssh
-rw-r--r--  1 yanto yanto    0 Jun  2 11:08 .sudo_as_admin_successful
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Templates
```

- d. Membuktikan file uptime telah tercopy


```

yanto@yanto-VirtualBox:~/Documents$ cp /proc//uptime /home/yanto
yanto@yanto-VirtualBox:~/Documents$ ls
File_Test.txt
yanto@yanto-VirtualBox:~/Documents$ ls -al /home/yanto
total 92
drwxr-x--- 16 yanto yanto 4096 Jun 18 07:00 .
drwxr-xr-x  3 root  root  4096 Mei 19 12:22 ..
-rw-----  1 yanto yanto   25 Jun  7 17:37 .bash_history
-rw-r--r--  1 yanto yanto  220 Mei 19 12:22 .bash_logout
-rw-r--r--  1 yanto yanto 3771 Mei 19 12:22 .bashrc
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .cache
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .config
-r--r--r--  1 yanto yanto 3500 Jun 18 06:58 cpuinfo
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Desktop
drwxr-xr-x  2 yanto yanto 4096 Jun 18 06:50 Documents
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Downloads
-rw-rw-r--  1 yanto yanto   32 Jun 18 06:53 File_Test.txt
drwx-----  2 yanto yanto 4096 Jun 18 06:13 .gnupg
drwx-----  3 yanto yanto 4096 Mei 19 12:51 .local
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Music
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Pictures
-rw-r--r--  1 yanto yanto  807 Mei 19 12:22 .profile
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Public
drwx-----  4 yanto yanto 4096 Mei 19 12:51 snap
drwx-----  2 yanto yanto 4096 Jun  7 17:01 .ssh
-rw-r--r--  1 yanto yanto    0 Jun  2 11:08 .sudo_as_admin_successful
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Templates

```

- e. Menghapus file uptime

```

yanto@yanto-VirtualBox:~$ rm /home/yanto/uptime
rm: remove write-protected regular file '/home/yanto/uptime'?
yanto@yanto-VirtualBox:~$

```

- f. Membuktikan file uptime telah dihapus

```
yanto@yanto-VirtualBox:~$ ls -al /home/yanto
total 92
drwxr-x--- 16 yanto yanto 4096 Jun 18 07:00 .
drwxr-xr-x  3 root  root  4096 Mei 19 12:22 ..
-rw-----  1 yanto yanto   25 Jun  7 17:37 .bash_history
-rw-r--r--  1 yanto yanto  220 Mei 19 12:22 .bash_logout
-rw-r--r--  1 yanto yanto 3771 Mei 19 12:22 .bashrc
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .cache
drwx----- 12 yanto yanto 4096 Jun 18 06:14 .config
-r--r--r--  1 yanto yanto 3500 Jun 18 06:58 cpuinfo
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Desktop
drwxr-xr-x  2 yanto yanto 4096 Jun 18 06:50 Documents
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Downloads
-rw-rw-r--  1 yanto yanto   32 Jun 18 06:53 File_Test.txt
drwx-----  2 yanto yanto 4096 Jun 18 06:13 .gnupg
drwx-----  3 yanto yanto 4096 Mei 19 12:51 .local
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Music
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Pictures
-rw-r--r--  1 yanto yanto  807 Mei 19 12:22 .profile
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Public
drwx-----  4 yanto yanto 4096 Mei 19 12:51 snap
drwx-----  2 yanto yanto 4096 Jun  7 17:01 .ssh
-rw-r--r--  1 yanto yanto    0 Jun  2 11:08 .sudo_as_admin_successful
drwxr-xr-x  2 yanto yanto 4096 Mei 19 12:51 Templates
```

- g. Rename file cpuinfo di folder home Anda menjadi infocpu

```
yanto@yanto-VirtualBox:~$ ls
cpuinfo  Documents  File_Test.txt  Pictures  snap      uptime
Desktop  Downloads  Music          Public    Templates Videos
yanto@yanto-VirtualBox:~$ mv File_Test.txt File_RenameTest.txt
yanto@yanto-VirtualBox:~$ ls
cpuinfo  Documents  File_RenameTest.txt  Pictures  snap      uptime
Desktop  Downloads  Music                Public    Templates Videos
yanto@yanto-VirtualBox:~$
```

7. Membuat Folder

- a. Membuat folder dengan nama NIM

```
yanto@yanto-VirtualBox:~$ mkdir 2311104059
yanto@yanto-VirtualBox:~$ ls
2311104059  Documents  Music  snap  Videos
cpuinfo     Downloads  Pictures  Templates
Desktop     File_RenameTest.txt  Public  uptime
yanto@yanto-VirtualBox:~$
```

- b. Membuat folder nama di dalam folder NIM

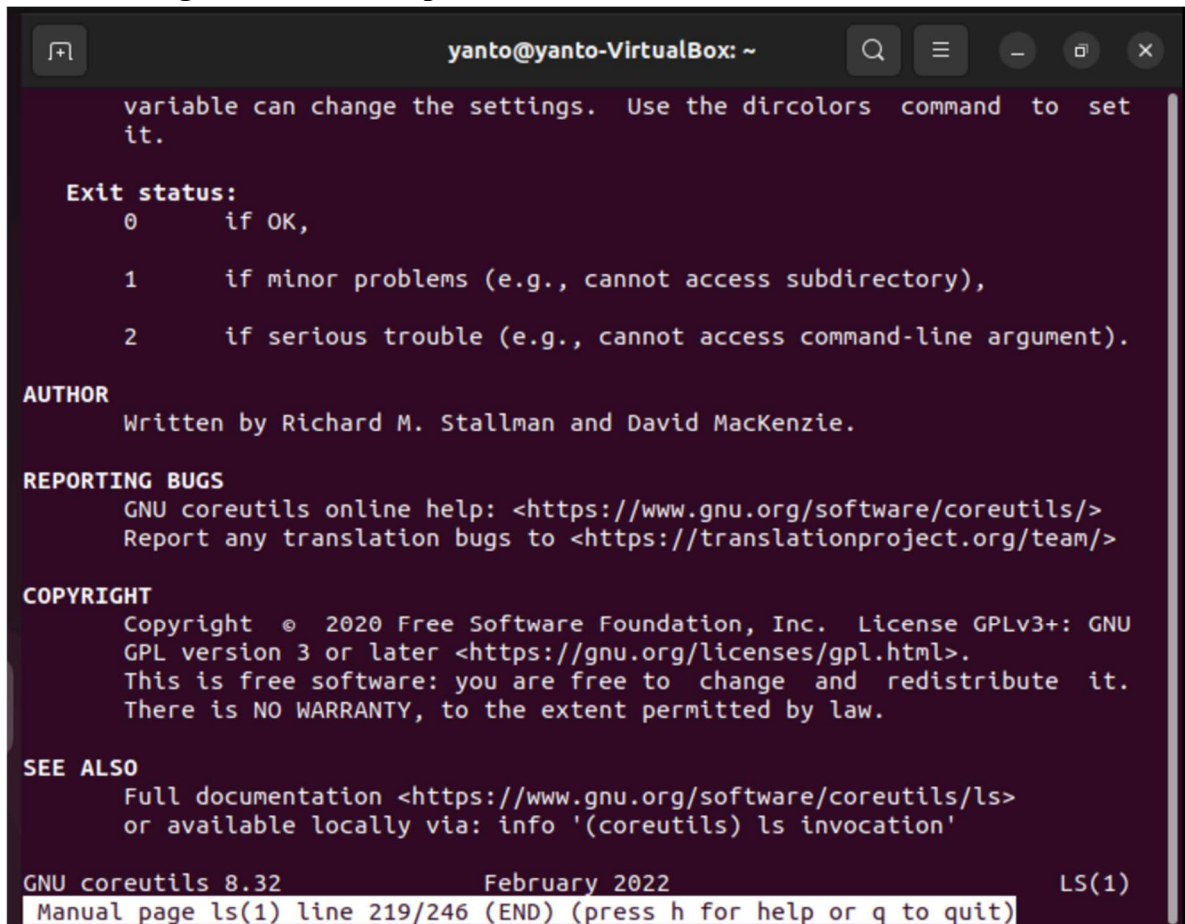
```

yanto@yanto-VirtualBox:~$ ls
2311104059  Documents      Music      snap      Videos
cpuinfo     Downloads      Pictures    Templates
Desktop     File_RenameTest.txt  Public     uptime
yanto@yanto-VirtualBox:~$ cd 2311104059
yanto@yanto-VirtualBox:~/2311104059$ mkdir ZhafirZaidanAvail
yanto@yanto-VirtualBox:~/2311104059$ ls
ZhafirZaidanAvail
yanto@yanto-VirtualBox:~/2311104059$

```

8. Membaca manual

a. Bukalah fungsi manual untuk perintah ls



```

yanto@yanto-VirtualBox: ~
variable can change the settings. Use the dircolors command to set
it.

Exit status:
0      if OK,

1      if minor problems (e.g., cannot access subdirectory),

2      if serious trouble (e.g., cannot access command-line argument).

AUTHOR
Written by Richard M. Stallman and David MacKenzie.

REPORTING BUGS
GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Report any translation bugs to <https://translationproject.org/team/>

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GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

SEE ALSO
Full documentation <https://www.gnu.org/software/coreutils/ls>
or available locally via: info '(coreutils) ls invocation'

GNU coreutils 8.32      February 2022      LS(1)
Manual page ls(1) line 219/246 (END) (press h for help or q to quit)

```

b. Apa fungsi perintah ls?

Perintah ls digunakan untuk menampilkan isi direktori atau file yang terdapat pada direktori aktif.

c. Siapakah pencipta perintah ls?

- Richard M. Stallman
- David MacKenzie

d. Apakah arti **-h** pada manual **ls**?

human-readable

Ukuran file ditampilkan dalam format yang mudah dibaca manusia, misalnya:

1K

20M

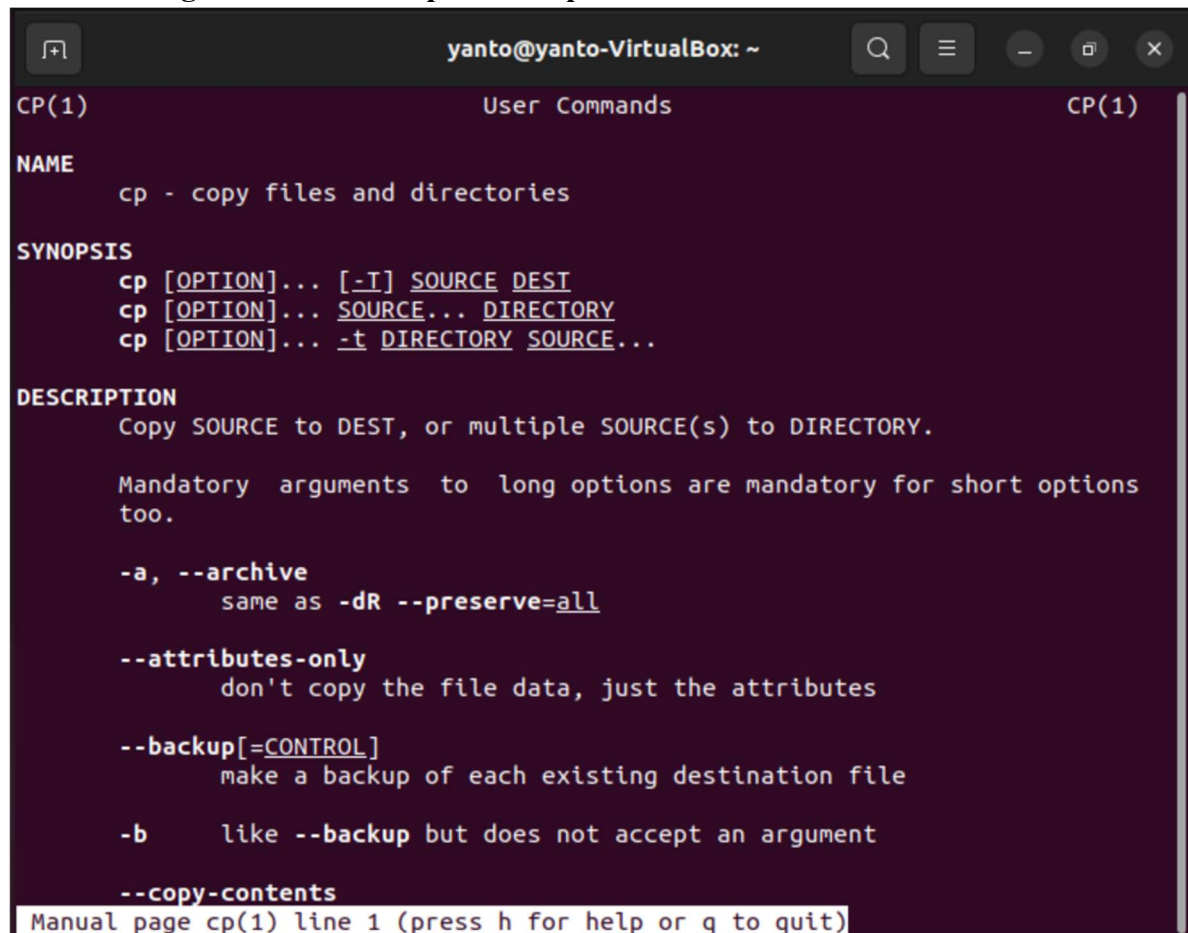
3G

e. Option apa yang digunakan agar dapat melihat direktori secara rekursif?

```
-R, --recursive
      list subdirectories recursively
```

Opsi **-R** akan menampilkan isi direktori beserta seluruh subdirektori secara rekursif.

f. Bukalah fungsi manual untuk perintah **cp**



```
yanto@yanto-VirtualBox: ~
CP(1)                                User Commands                                CP(1)

NAME
    cp - copy files and directories

SYNOPSIS
    cp [OPTION]... [-I] SOURCE DEST
    cp [OPTION]... SOURCE... DIRECTORY
    cp [OPTION]... -t DIRECTORY SOURCE...

DESCRIPTION
    Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --archive
        same as -dR --preserve=all

    --attributes-only
        don't copy the file data, just the attributes

    --backup[=CONTROL]
        make a backup of each existing destination file

    -b
        like --backup but does not accept an argument

    --copy-contents
    Manual page cp(1) line 1 (press h for help or q to quit)
```

g. Apa fungsi perintah **cp**?

Menyalin (copy) file atau direktori dari suatu lokasi ke lokasi lain.

h. Siapakah pencipta perintah cp?

AUTHOR

Written by Torbjorn Granlund, David MacKenzie, and Jim Meyering.

i. Apakah arti -v pada perintah cp?

-v, --verbose
explain what is being done

verbose

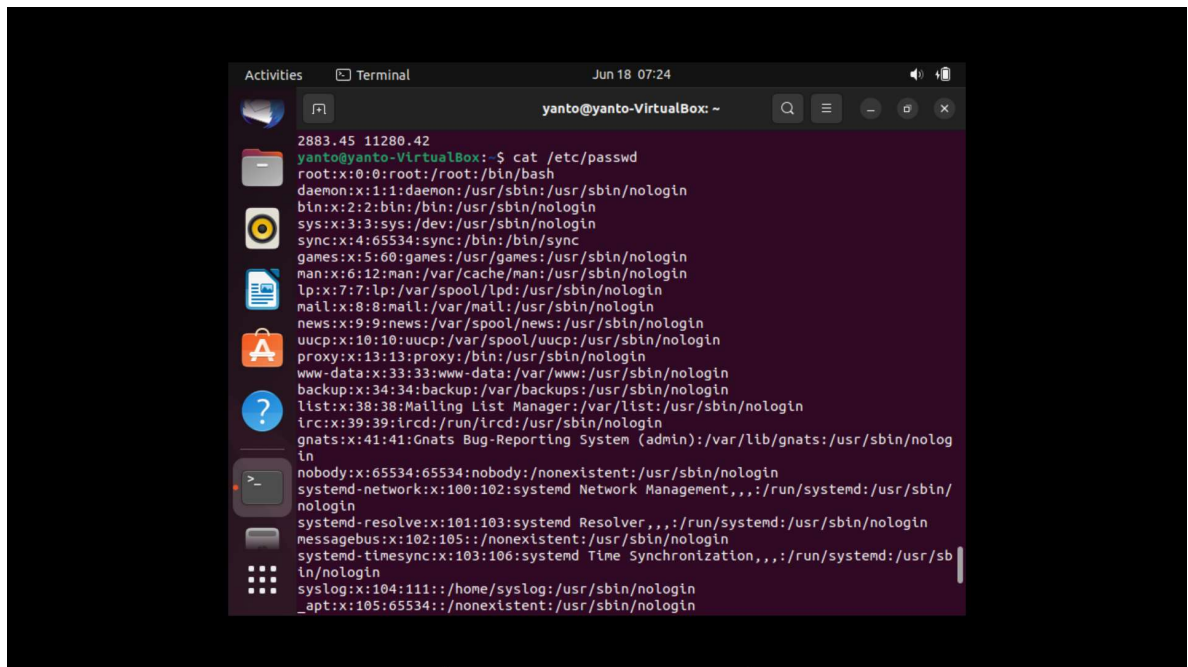
Perintah akan menampilkan file yang sedang disalin.

j. Jika ingin interaktif, option apa yang harus digunakan?

k.

9. Pipe

a. Lakukan perintah ini `cat /etc/passwd` dan screenshot hasil perintah tersebut!



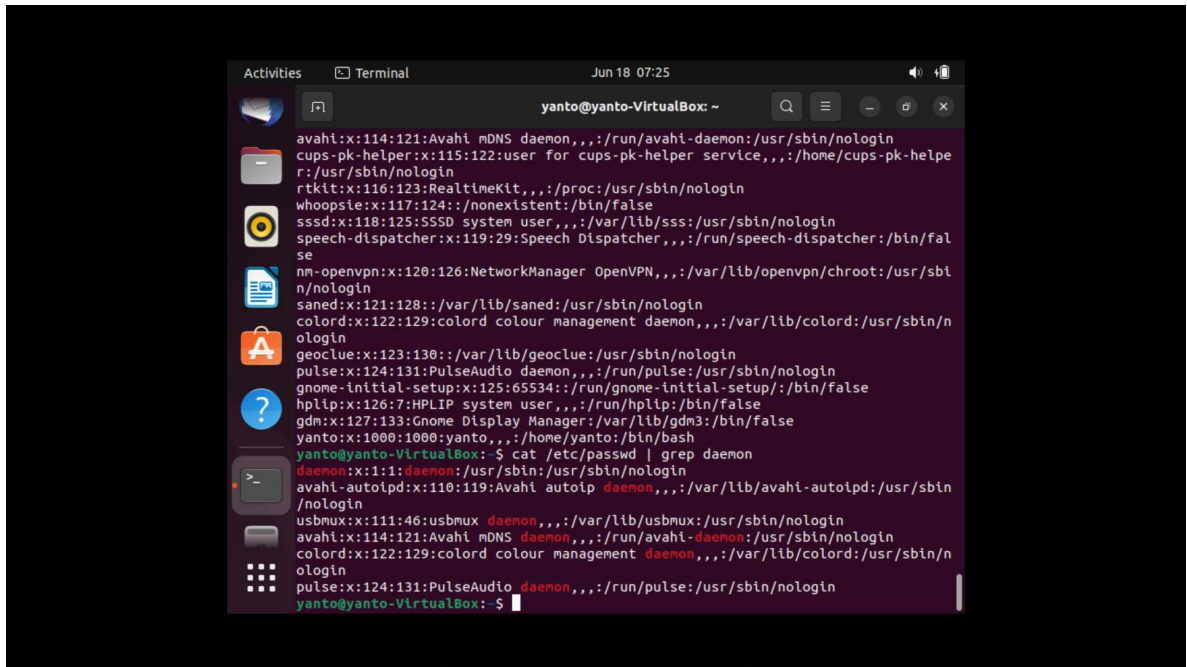
```
2883.45 11280.42
yanto@yanto-VirtualBox: ~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:MailList Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:,:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:,:/home/syslog:/usr/sbin/nologin
_apt:x:105:65534:,:/nonexistent:/usr/sbin/nologin
```

b. Apa fungsi perintah cat?

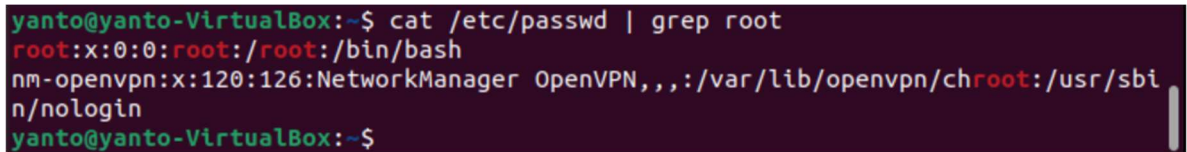
Perintah cat (concatenate) digunakan untuk:

- Menampilkan isi file ke terminal.
- Menggabungkan beberapa file.
- Membuat file baru dengan redirection.

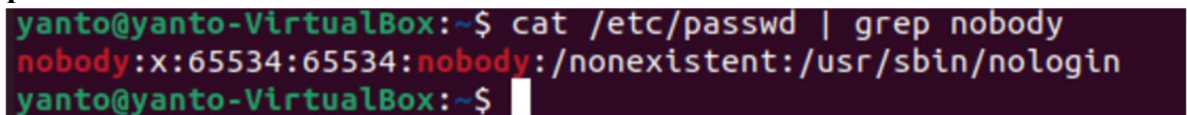
- c. Lakukan perintah `cat /etc/passwd | grep daemon` dan screenshot hasil perintah tersebut!



- d. Lakukan perintah `cat /etc/passwd | grep root` dan screenshot hasil perintah tersebut!



- e. Lakukan perintah `cat /etc/passwd | grep nobody` dan screenshot hasil perintah tersebut!



- f. Apakah fungsi perintah “`|` grep daemon”?
Simbol:

|
adalah pipe, yang berfungsi meneruskan output dari perintah sebelumnya menjadi input bagi perintah berikutnya.

Sedangkan:

grep daemon

berfungsi mencari dan menampilkan baris yang mengandung kata daemon.

Jadi:

cat /etc/passwd | grep daemon

berfungsi untuk:

Menampilkan hanya baris pada file /etc/passwd yang mengandung kata daemon.

10. Redirection

a. Jalankan perintah berikut dan jelaskan hasilnya

```
yanto@yanto-VirtualBox:~$ cd /
yanto@yanto-VirtualBox:/$ ls -al > /home/yanto/result.txt
yanto@yanto-VirtualBox:/$ cat /home/yanto/result.txt
total 1095768
drwxr-xr-x 20 root root      4096 Mei 19 12:21 .
drwxr-xr-x 20 root root      4096 Mei 19 12:21 ..
lrwxrwxrwx  1 root root         7 Mei 19 12:12 bin -> usr/bin
drwxr-xr-x  4 root root      4096 Mei 19 12:31 boot
drwxrwxr-x  2 root root      4096 Mei 19 12:21 cdrom
drwxr-xr-x 19 root root     4100 Jun 18 06:12 dev
drwxr-xr-x 129 root root    12288 Jun  2 11:08 etc
drwxr-xr-x  3 root root      4096 Mei 19 12:22 home
lrwxrwxrwx  1 root root         7 Mei 19 12:12 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Mei 19 12:12 lib32 -> usr/lib32
lrwxrwxrwx  1 root root         9 Mei 19 12:12 lib64 -> usr/lib64
lrwxrwxrwx  1 root root        10 Mei 19 12:12 libx32 -> usr/libx32
drwx----- 2 root root    16384 Mei 19 12:12 lost+found
drwxr-xr-x  2 root root      4096 Agu  9 2022 media
drwxr-xr-x  2 root root      4096 Agu  9 2022 mnt
drwxr-xr-x  2 root root      4096 Agu  9 2022 opt
dr-xr-xr-x 262 root root         0 Jun 18 06:12 proc
drwx----- 4 root root      4096 Jun  2 11:04 root
drwxr-xr-x 33 root root       880 Jun 18 06:34 run
lrwxrwxrwx  1 root root         8 Mei 19 12:12/sbin -> usr/sbin
drwxr-xr-x 11 root root      4096 Agu  9 2022 snap
drwxr-xr-x  2 root root      4096 Agu  9 2022 srv
```

b. Di mana file result.txt berada?

```
yanto@yanto-VirtualBox:~$ ls
2311104059 Documents Music result.txt uptime
cpuinfo Downloads Pictures snap Videos
Desktop File_RenameTest.txt Public Templates
```

c. Jalankan perintah berikut dan jelaskan hasilnya


```

yanto@yanto-VirtualBox:~$ cd /etc
yanto@yanto-VirtualBox:/etc$ ls -al > /home/yanto/result.txt
yanto@yanto-VirtualBox:/etc$ cat /home/yanto/result.txt
total 1136
drwxr-xr-x 129 root root 12288 Jun  2 11:08 .
drwxr-xr-x 20 root root 4096 Mei 19 12:21 ..
drwxr-xr-x  3 root root 4096 Agu  9 2022 acpi
-rw-r--r--  1 root root 3028 Agu  9 2022 adduser.conf
drwxr-xr-x  3 root root 4096 Agu  9 2022 alsa
drwxr-xr-x  2 root root 4096 Mei 19 12:26 alternatives
-rw-r--r--  1 root root  335 Mar 23 2022 anacrontab
-rw-r--r--  1 root root  433 Mar 23 2022 apg.conf
drwxr-xr-x  5 root root 4096 Agu  9 2022 apm
drwxr-xr-x  3 root root 4096 Agu  9 2022 apparmor
drwxr-xr-x  7 root root 4096 Mei 19 12:31 apparmor.d
drwxr-xr-x  4 root root 4096 Agu  9 2022 apport
-rw-r--r--  1 root root  769 Feb 22 2022 appstream.conf
drwxr-xr-x  8 root root 4096 Mei 19 12:32 apt
drwxr-xr-x  3 root root 4096 Agu  9 2022 avahi
-rw-r--r--  1 root root 2319 Jan  6 2022 bash.bashrc
-rw-r--r--  1 root root  45 Nov 11 2021 bash_completion
drwxr-xr-x  2 root root 4096 Agu  9 2022 bash_completion.d
-rw-r--r--  1 root root  367 Des 16 2020 bindresvport.blacklist
drwxr-xr-x  2 root root 4096 Apr  8 2022 binfmt.d
drwxr-xr-x  2 root root 4096 Agu  9 2022 bluetooth
-rw-r----- 1 root root  33 Agu  9 2022 brlapi.key
drwxr-xr-x  7 root root 4096 Agu  9 2022 brltty

```

- d. Apakah fungsi dari perintah >?
Mengarahkan output suatu perintah ke file dan menimpa isi file sebelumnya (overwrite).
- e. Jalankan perintah berikut dan jelaskan hasilnya

```

yanto@yanto-VirtualBox:/$ cat /home/yanto/result1.txt
total 1095768
drwxr-xr-x 20 root root      4096 Mei 19 12:21 .
drwxr-xr-x 20 root root      4096 Mei 19 12:21 ..
lrwxrwxrwx 1 root root         7 Mei 19 12:12 bin -> usr/bin
drwxr-xr-x 4 root root      4096 Mei 19 12:31 boot
drwxrwxr-x 2 root root      4096 Mei 19 12:21 cdrom
drwxr-xr-x 19 root root     4100 Jun 18 06:12 dev
drwxr-xr-x 129 root root    12288 Jun  2 11:08 etc
drwxr-xr-x 3 root root      4096 Mei 19 12:22 home
lrwxrwxrwx 1 root root         7 Mei 19 12:12 lib -> usr/lib
lrwxrwxrwx 1 root root         9 Mei 19 12:12 lib32 -> usr/lib32
lrwxrwxrwx 1 root root         9 Mei 19 12:12 lib64 -> usr/lib64
lrwxrwxrwx 1 root root        10 Mei 19 12:12 libx32 -> usr/libx32
drwx----- 2 root root    16384 Mei 19 12:12 lost+found
drwxr-xr-x 2 root root      4096 Agu  9 2022 media
drwxr-xr-x 2 root root      4096 Agu  9 2022 mnt
drwxr-xr-x 2 root root      4096 Agu  9 2022 opt
dr-xr-xr-x 262 root root      0 Jun 18 06:12 proc
drwx----- 4 root root      4096 Jun  2 11:04 root
drwxr-xr-x 33 root root      880 Jun 18 06:34 run
lrwxrwxrwx 1 root root         8 Mei 19 12:12 sbin -> usr/sbin
drwxr-xr-x 11 root root      4096 Agu  9 2022 snap
drwxr-xr-x 2 root root      4096 Agu  9 2022 srv
-rw----- 1 root root 1121976320 Mei 19 12:12 swapfile
dr-xr-xr-x 13 root root      0 Jun 18 06:12 sys
drwxrwxrwt 19 root root      4096 Jun 18 06:41 tmp

```

f. Jalankan perintah berikut dan jelaskan hasilnya

```

yanto@yanto-VirtualBox:/etc$ ls -al >> /home/yanto/result1.txt
yanto@yanto-VirtualBox:/etc$ cat /home/yanto/result1.txt
total 1095768
drwxr-xr-x 20 root root      4096 Mei 19 12:21 .
drwxr-xr-x 20 root root      4096 Mei 19 12:21 ..
lrwxrwxrwx 1 root root         7 Mei 19 12:12 bin -> usr/bin
drwxr-xr-x 4 root root      4096 Mei 19 12:31 boot
drwxrwxr-x 2 root root      4096 Mei 19 12:21 cdrom
drwxr-xr-x 19 root root     4100 Jun 18 06:12 dev
drwxr-xr-x 129 root root    12288 Jun  2 11:08 etc
drwxr-xr-x 3 root root      4096 Mei 19 12:22 home
lrwxrwxrwx 1 root root         7 Mei 19 12:12 lib -> usr/lib
lrwxrwxrwx 1 root root         9 Mei 19 12:12 lib32 -> usr/lib32
lrwxrwxrwx 1 root root         9 Mei 19 12:12 lib64 -> usr/lib64
lrwxrwxrwx 1 root root        10 Mei 19 12:12 libx32 -> usr/libx32
drwx----- 2 root root    16384 Mei 19 12:12 lost+found
drwxr-xr-x 2 root root      4096 Agu  9 2022 media
drwxr-xr-x 2 root root      4096 Agu  9 2022 mnt
drwxr-xr-x 2 root root      4096 Agu  9 2022 opt
dr-xr-xr-x 262 root root      0 Jun 18 06:12 proc
drwx----- 4 root root      4096 Jun  2 11:04 root
drwxr-xr-x 33 root root      880 Jun 18 06:34 run
lrwxrwxrwx 1 root root         8 Mei 19 12:12 sbin -> usr/sbin
drwxr-xr-x 11 root root      4096 Agu  9 2022 snap
drwxr-xr-x 2 root root      4096 Agu  9 2022 srv
-rw----- 1 root root 1121976320 Mei 19 12:12 swapfile
dr-xr-xr-x 13 root root      0 Jun 18 06:12 sys

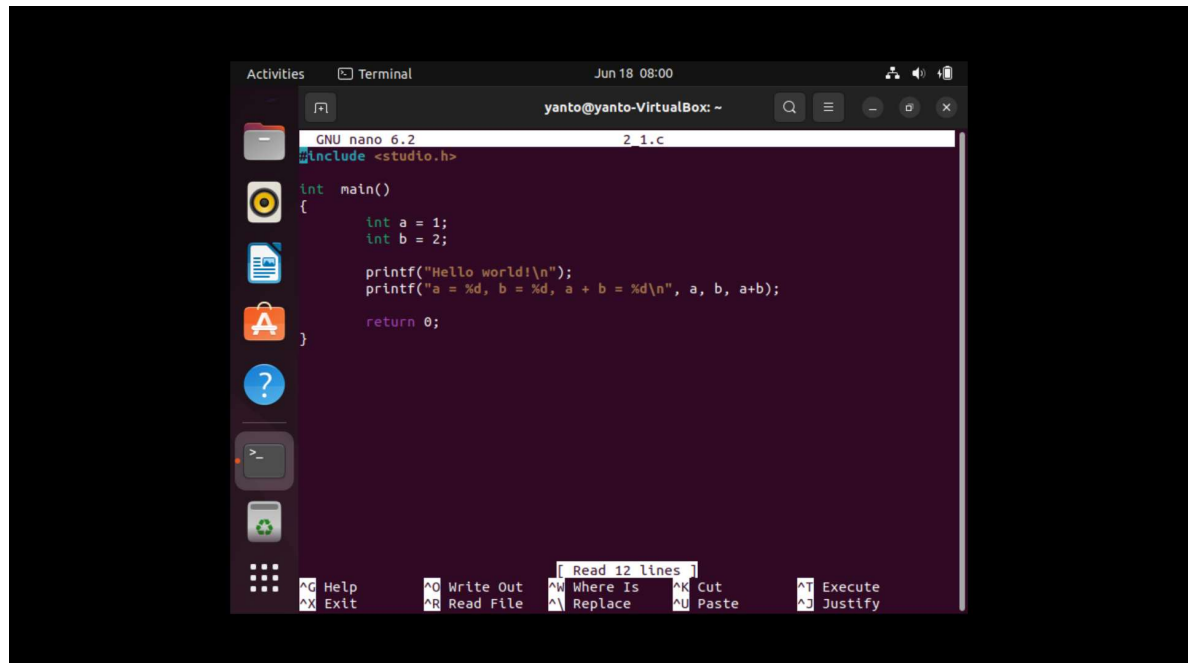
```

g. Apakah perbedaan perintah > dan >>?

Simbol	Fungsi
>	Menyimpan output ke file dan menimpa isi sebelumnya (overwrite)
>>	Menambahkan output ke akhir file tanpa menghapus isi sebelumnya (append)

11. Compile Source Code

a. Membuat file 2_1.c



```
GNU nano 6.2 2_1.c
#include <stdio.h>

int main()
{
    int a = 1;
    int b = 2;

    printf("Hello world!\n");
    printf("a = %d, b = %d, a + b = %d\n", a, b, a+b);

    return 0;
}
```

b. Kompilasi source code menjadi program bernama 2_1

A terminal window titled 'yanto@yanto-VirtualBox: ~' showing the process of compiling a C program. The user runs 'sudo apt install <deb name>', 'cd ~', 'nano 2_1.c', and 'gcc -o 2_1 2_1.c'. The compilation fails with the error 'fatal error: studio.h: No such file or directory'. The user then runs 'ls' and 'gcc -o 2_1 2_1.c' again, but the error persists. The terminal output shows the error message and the compilation terminated.

```
command 'cdde' from deb cdde (0.3.1-1build1)
Try: sudo apt install <deb name>
yanto@yanto-VirtualBox:~$ cd ~
yanto@yanto-VirtualBox:~$ nano 2_1.c
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
2_1.c:1:10: fatal error: studio.h: No such file or directory
1 | #include <studio.h>
|
compilation terminated.
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
gcc: fatal error: no input files
compilation terminated.
yanto@yanto-VirtualBox:~$ ls
2_1.c      Desktop    File_RenameTest.txt  Public      snap        Videos
2311104059 Documents  Music                result1.txt Templates
cpuinfo    Downloads  Pictures              result.txt  uptime
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
2_1.c:1:10: fatal error: studio.h: No such file or directory
1 | #include <studio.h>
|
compilation terminated.
yanto@yanto-VirtualBox:~$ nano 2_1.c
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
yanto@yanto-VirtualBox:~$ ls
2_1      cpuinfo    Downloads  Pictures    result.txt  uptime
2_1.c    Desktop    File_RenameTest.txt  Public      snap        Videos
2311104059 Documents  Music                result1.txt Templates
yanto@yanto-VirtualBox:~$
```

c. Menjalankan program

A terminal window titled 'yanto@yanto-VirtualBox: ~' showing the execution of the program. The user runs 'nano 2_1.c', 'gcc -o 2_1 2_1.c', and './2_1'. The program outputs 'Hello world!' and 'a = 1, b = 2, a + b = 3'. The terminal output shows the error message and the compilation terminated.

```
yanto@yanto-VirtualBox:~$ nano 2_1.c
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
2_1.c:1:10: fatal error: studio.h: No such file or directory
1 | #include <studio.h>
|
compilation terminated.
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
gcc: fatal error: no input files
compilation terminated.
yanto@yanto-VirtualBox:~$ ls
2_1      cpuinfo    Downloads  Pictures    result.txt  uptime
2_1.c    Desktop    File_RenameTest.txt  Public      snap        Videos
2311104059 Documents  Music                result1.txt Templates
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
2_1.c:1:10: fatal error: studio.h: No such file or directory
1 | #include <studio.h>
|
compilation terminated.
yanto@yanto-VirtualBox:~$ nano 2_1.c
yanto@yanto-VirtualBox:~$ gcc -o 2_1 2_1.c
yanto@yanto-VirtualBox:~$ ls
2_1      cpuinfo    Downloads  Pictures    result.txt  uptime
2_1.c    Desktop    File_RenameTest.txt  Public      snap        Videos
2311104059 Documents  Music                result1.txt Templates
yanto@yanto-VirtualBox:~$ ./2_1
Hello world!
a = 1, b = 2, a + b = 3
yanto@yanto-VirtualBox:~$
```

d. Membuat file 2_2.c


```
GNU nano 6.2 2_2.c
#include <string.h>
#include <errno.h>

int main(int argc, char *argv[])
{
    int fd;
    if (2 != argc)
    {
        printf("\nPengumuman 'myopen 2_2.c'\n");
        return 1;
    }
    errno = 0;
    fd = open(argv[1], O_RDONLY);
    if (-1 == fd)
    {
        printf("\nmyopen gagal membuka file %s dengan pesan error:\n%s",
            argv[1], strerror(errno));
        return 1;
    }
    else
    {
        printf("myopen berhasil membuka file %s\n", argv[1]);
    }
}
```

- e. Compile menjadi program myopen

```
yanto@yanto-VirtualBox:~$ gcc -o myopen 2_2.c
yanto@yanto-VirtualBox:~$ ls
2_1      cpuinfo    File_RenameTest.txt  Public      Templates
2_1.c    Desktop    Music                 result1.txt uptime
2_2.c    Documents  myopen                result.txt  Videos
2311104059 Downloads  Pictures              snap
```

- f. Menjalankan program myopen

```
yanto@yanto-VirtualBox:~$ ./myopen

Pengumuman 'myopen 2_2.c'
```

- g. Jalankan dengan file yang ada

```
yanto@yanto-VirtualBox:~$ nano 2_2.c
yanto@yanto-VirtualBox:~$ ./myopen

Pengumuman 'myopen 2_2.c'
yanto@yanto-VirtualBox:~$ ./myopen /etc/passwd
myopen berhasil membuka file /etc/passwd
yanto@yanto-VirtualBox:~$
```

- h. Jelaskan fungsi program tersebut

Program myopen digunakan untuk membuka file yang diberikan melalui parameter command line. Jika file berhasil dibuka, program akan menampilkan pesan bahwa file berhasil dibuka. Jika file tidak ditemukan

atau gagal dibuka, program akan menampilkan pesan error sesuai penyebab kegagalannya.